TierZero

Self-Healing Adaptive Browser Automation

Record a workflow once. Replay it anywhere.

The UI changes. The workflow doesn't break.

Steve Elliott -- March 2026

The Problem: Brittle Automation

Every enterprise has repetitive browser workflows -- ServiceNow tickets, insurance claims, HR onboarding. Today they're either done **manually** or automated with **fragile scripts that break constantly**.

What breaks scripts:

- UI redesign changes CSS selectors
- Button text changes ("Save" to "Apply")
- New confirmation modal appears
- Table columns get reordered
- Form fields move to different section

The cost:

- **$50-400B/year** per vertical in manual services
- RPA implementations fail 30-50% of the time
- Every UI update = weeks of script maintenance
- Companies hire humans instead of fixing scripts

> **The gap:** Scripted automation = consulting gig (rebuild every time). Self-healing automation = scalable product.

The Solution: Intent-Based Automation

Instead of telling the computer **HOW** to interact (CSS selectors), tell it **WHAT** to achieve (goals). The system figures out the HOW at runtime.

Traditional (Scripted)

```
page.click('#btn-resolve-main')
```

```
page.fill('#comment-textarea', text)
```

```
page.click('button.submit-btn')
```

Selector changes = script dies. Every time.

TierZero (Intent-Based)

```
goal: "Click the Resolve button"
```

```
goal: "Fill the comment field"
```

```
goal: "Submit the form"
```

UI changes = system adapts automatically.

Key insight: The workflow describes the GOAL ("click Resolve"), not the SELECTOR ("#btn-resolve"). When the UI changes, the IntentEngine finds the element by *understanding the page*, not memorizing DOM paths.

The Full Pipeline

From human demonstration to adaptive automated skill in 5 steps:

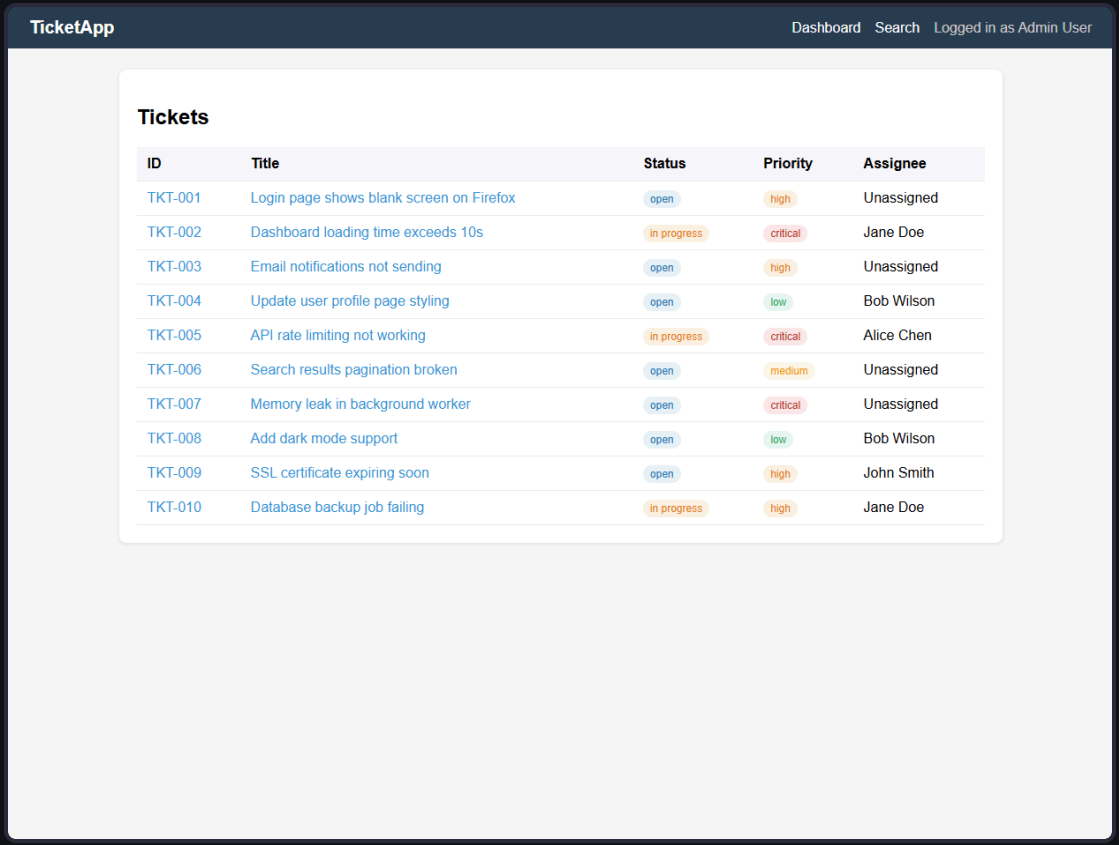
🎥 Record
Watch human do task
→
🏷️ Annotate
LLM adds meaning
→
⚙️ Generate
Create intent workflow
→
📦 Package
Hot-loadable skill
→
▶️ Replay
Adaptive execution

- **Record:** CDP captures every click, keystroke, navigation with page state snapshots
- **Annotate:** LLM generates semantic descriptions and detects variable vs constant values
- **Generate:** Converts recording into intent-based workflow steps -- goals, not selectors
- **Package:** Outputs a hot-loadable TierZero skill with manifest and executable entry point
- **Replay:** IntentEngine executes each intent adaptively with 5-strategy fallback + recovery

> **No coding required.** A business analyst demonstrates the task. TierZero generates the automation. If the UI changes, it adapts.

Demo: Layout v1 -- Dashboard (Original Design)

A simulated ticket management system. 10 tickets. Standard layout: ID first, status/priority inline, nav right-aligned. **This is where we record the workflow.**

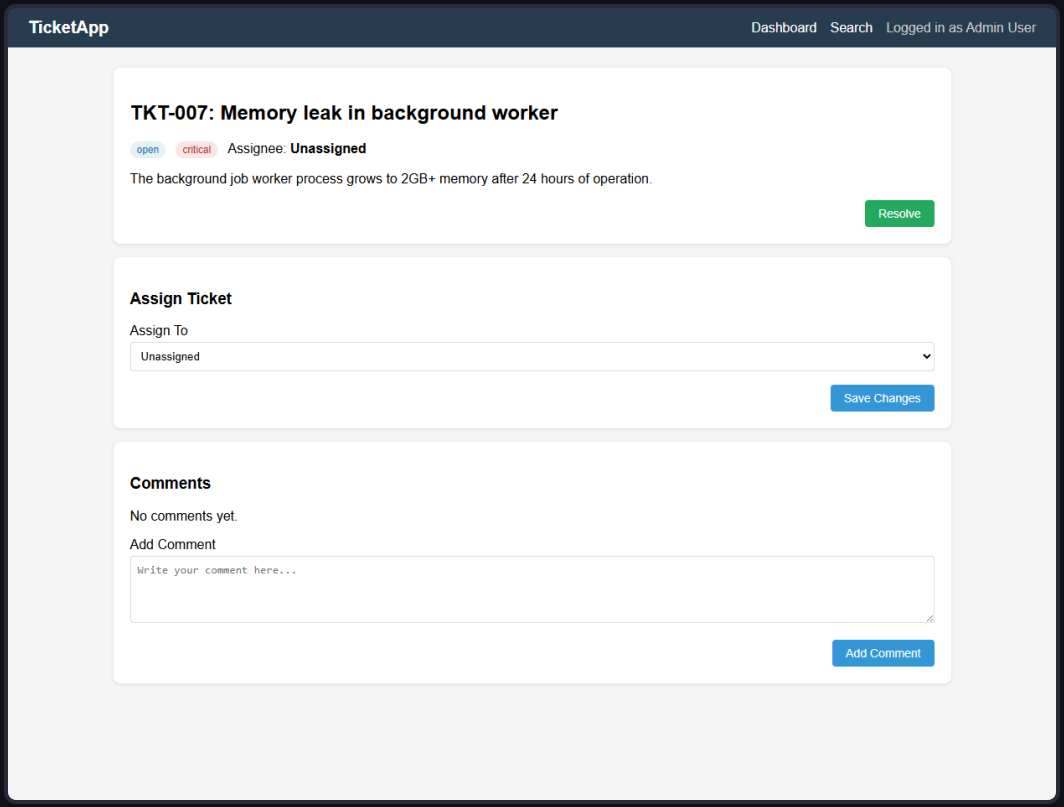


ID	Title	Status	Priority	Assignee
TKT-001	Login page shows blank screen on Firefox	open	high	Unassigned
TKT-002	Dashboard loading time exceeds 10s	in progress	critical	Jane Doe
TKT-003	Email notifications not sending	open	high	Unassigned
TKT-004	Update user profile page styling	open	low	Bob Wilson
TKT-005	API rate limiting not working	in progress	critical	Alice Chen
TKT-006	Search results pagination broken	open	medium	Unassigned
TKT-007	Memory leak in background worker	open	critical	Unassigned
TKT-008	Add dark mode support	open	low	Bob Wilson
TKT-009	SSL certificate expiring soon	open	high	John Smith
TKT-010	Database backup job failing	in progress	high	Jane Doe

Column order: ID → Title → Status → Priority → Assignee. Blue accent. Buttons in nav bar.

Layout v1 -- Ticket Detail Page

The recorded workflow: open TKT-007, add comment, assign, click Resolve. All buttons are **text-based** and **right-aligned**.



"Resolve" (green, bottom-right). **"Save Changes"** (blue, right). **"Add Comment"** (blue, right). Standard text labels on all buttons.

Now: The UI Gets Redesigned

Product team ships a redesign. **Traditional automation breaks immediately.** Here's what changed:

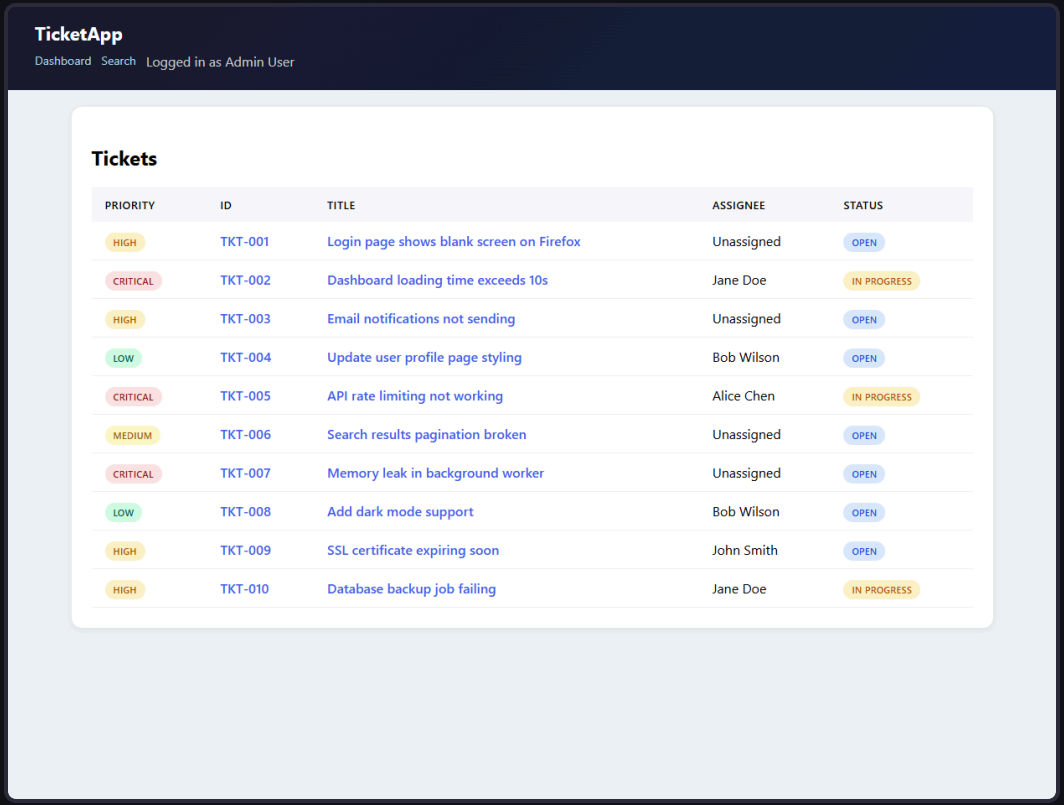
- **Button text changed:** "Resolve" → "✓ Done" "Add Comment" → "💬 Comment" "Save Changes" → "💾 Save"
- **Button position changed:** Moved from bottom-right to top-left and bottom-left
- **CSS classes changed:** `.btn` → `.action-btn`, `.card` → `.panel`
- **Table columns reordered:** Priority moved to FIRST column, Status to LAST
- **Labels uppercased:** "Username" → "USERNAME", "Add Comment" → "ADD COMMENT"
- **Color scheme changed:** Light blue → Purple/Indigo
- **New confirmation modal:** v2 shows "Are you sure?" on resolve (didn't exist before)

With scripted automation: Every selector fails. Every text match fails. Script is dead. Engineer spends days rewriting.

With TierZero: Workflow says "Click Resolve button." IntentEngine finds "✓ Done" via aria-label. Zero changes needed.

Layout v2 -- Redesigned Dashboard

Same data. **Completely different structure.** Priority column FIRST. Headers UPPERCASE. Status pills on RIGHT. Nav below logo. Different background color.

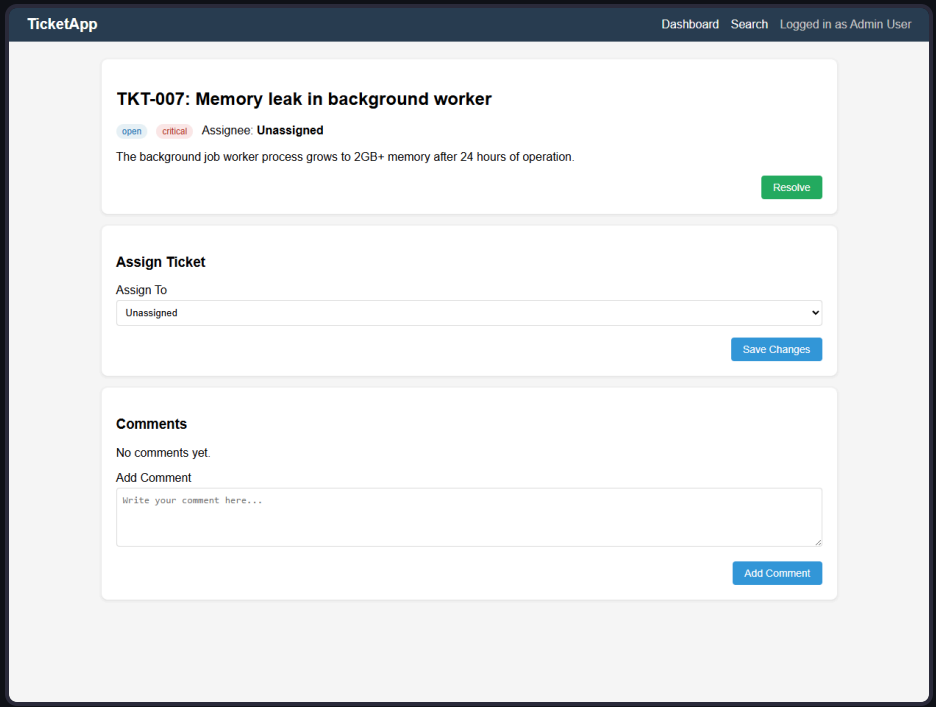


Every CSS selector is different. Column order: Priority → ID → Title → Assignee → Status. A scripted automation would fail on every single element.

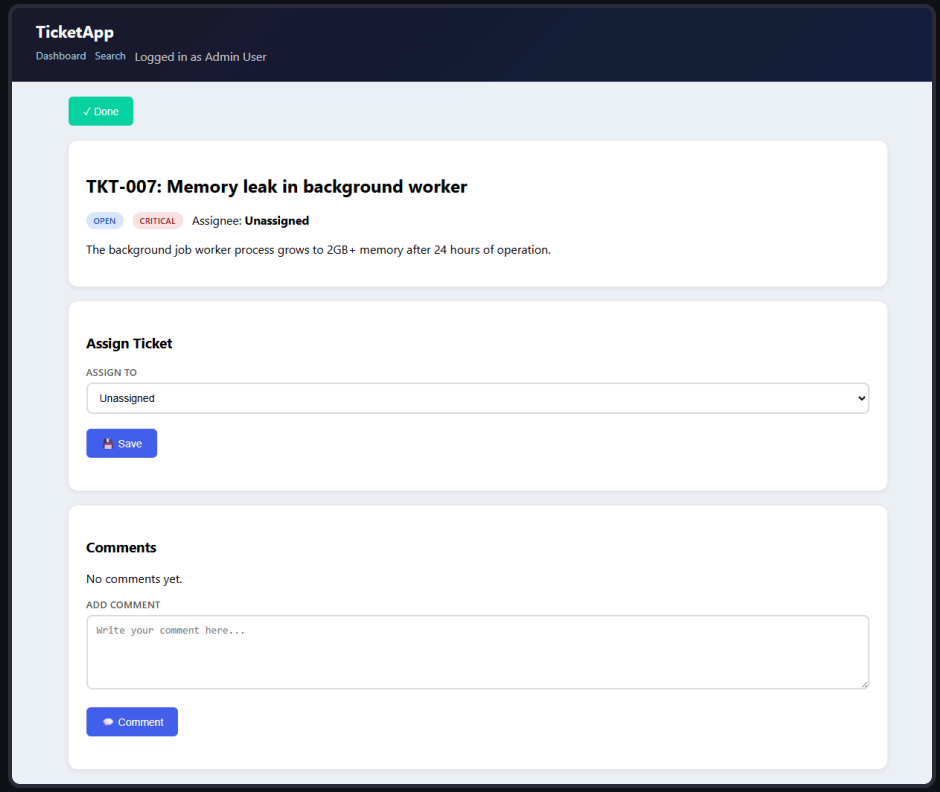
Side-by-Side: The Adaptive Challenge

Workflow recorded on v1 (left). Replayed on v2 (right). **Zero modifications to the workflow.**

v1 -- Workflow Recorded Here



v2 -- Replayed Successfully ✓



Every difference handled: "Resolve" → "✓ Done" (found via aria-label). Button moved top-left (found via role). New confirm modal auto-dismissed. Comment field found despite uppercase label change.

How the IntentEngine Adapts

Five resolution strategies tried in order. If one fails, the next kicks in automatically:

- **1. Cached (~50ms):** Try last-known-good selector. If UI hasn't changed, instant.
- **2. Aria (~200ms):** Find by accessibility role + label. "Resolve" found even when text is "✓ Done" because `aria-label` is preserved.
- **3. LLM -- Accessibility Tree (~500ms):** Feed page's a11y tree to GPT-4o-mini. Understands context: "this looks like a resolve/submit action."
- **4. LLM -- Vision (~1s):** Screenshot + GPT-4o vision. Finds anything visible on screen.
- **5. Coordinates (~1.5s):** Ask LLM for pixel coordinates, click directly. Last resort.

Recovery strategies:

- **Dialog Dismissal:** Detects and closes unexpected modals/alerts automatically
- **LLM Recovery:** Analyzes page state, suggests navigation or wait actions
- **Page State Assertions:** Verifies expected state between steps, retries if wrong

> The system **never just crashes**. It tries 5 strategies, then 3 recovery methods, then escalates with a clear explanation of what failed.

Real-World Applications

IT Service Management

Auto-resolve ServiceNow/Jira tickets. Investigate alerts in App Insights. Update status, add comments, assign owners.

Insurance/Gov

Process claims, renewals, cancellations across legacy web portals. Handles form changes gracefully.

Any Enterprise

Onboarding, procurement, compliance checks -- any repetitive browser task a human does today.

The pitch: "Show me how you do it once. I'll do it forever, even when the UI changes."

> **Sequoia/Julien Bek framing:** Sell outcomes ("your queue is zero by morning"), not tools. Autopilot, not copilot.

Built in One Sunday Morning

9,605	243	18	0
lines of code	tests passing	new modules	failures

Scripted automation = **consulting gig**

Self-healing adaptive automation = **scalable business**

$50-400B TAM per vertical in services automation.